

27.1 Basic configuration

The RTC is configured using the following registers:

1. Power: In the PCONP register ([Table 46](#)), set bits PCRTC.

Remark: On reset, the RTC is enabled. See [Section 27.7](#) for power saving options.

2. Clock: The RTC uses the 1 Hz clock output from the RTC oscillator as the only clock source. The peripheral clock rate for accessing registers is CCLK/8.
3. Interrupts: See [Section 27.6.1](#) for RTC interrupt handling. Interrupts are enabled in the NVIC using the appropriate Interrupt Set Enable register. The RTC interrupt can wake up the device from deep-sleep, power-down, or deep power-down modes. See [Section 27.6.1](#).

27.2 Features

- Measures the passage of time to maintain a calendar and clock. Provides seconds, minutes, hours, day of month, month, year, day of week, and day of year.
- Ultra-low power design to support battery powered systems. Less than 1 microamp required for battery operation. Uses power from the CPU power supply when it is present.
- 20 bytes of Battery-backed storage and RTC operation when power is removed from the CPU.
- Dedicated 32 kHz ultra low power oscillator.
- Dedicated battery power supply pin.
- RTC power supply is isolated from the rest of the chip.
- Calibration counter allows adjustment to better than ± 1 sec/day with 1 sec resolution.
- Periodic interrupts can be generated from increments of any field of the time registers.
- Alarm interrupt can be generated for a specific date/time.

27.3 Description

The Real Time Clock (RTC) is a set of counters for measuring time when system power is on, and optionally when it is off. It uses very little power when its registers are not being accessed by the CPU, especially reduced power modes. On the LPC176x/5x, the RTC is clocked by a separate 32 kHz oscillator that produces a 1 Hz internal time reference. The RTC is powered by its own power supply pin, VBAT, which can be connected to a battery, externally tied to a 3V supply, or left floating.

The RTC power domain is shown in conceptual form in [Figure 131](#). A detailed view of the time keeping portion of the RTC is shown in [Figure 132](#).

27.4 Architecture

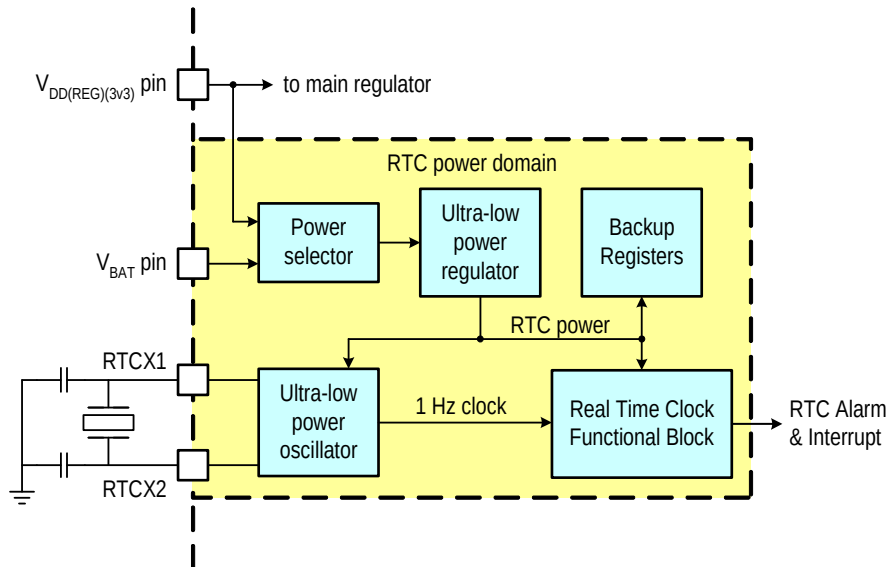


Fig 131. RTC domain conceptual diagram

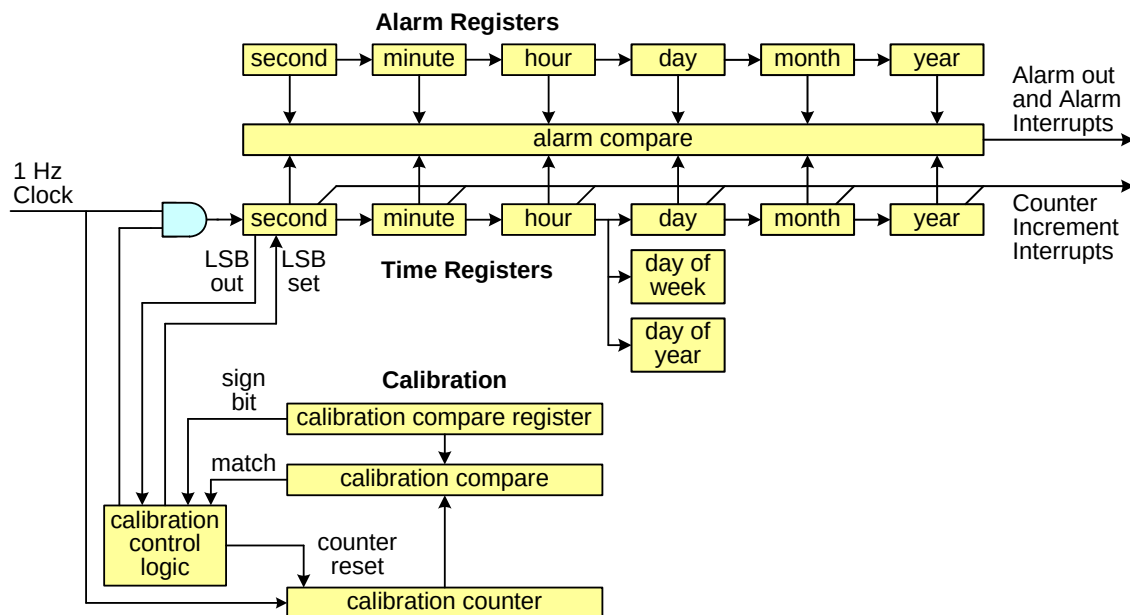


Fig 132. RTC functional block diagram

27.5 Pin description

Table 507. RTC pin description

Name	Type	Description
RTCX1	I	Input to the RTC oscillator circuit.
RTCX2	O	Output from the RTC oscillator circuit. Remark: If the RTC is not used, the RTCX1/2 pins can be left floating.
V _{BAT}	I	RTC power supply: Typically connected to an external 3V battery. If this pin is not powered, the RTC is still powered internally if V _{DD(REG)(3V3)} is present.

27.6 Register description

The RTC includes a number of registers, shown in [Table 508](#). Detailed descriptions of the registers follow. In these descriptions, for most of the registers the Reset Value column shows "NC", meaning that these registers are Not Changed by a Reset. Software must initialize these registers between power-on and setting the RTC into operation. The registers are split into five sections by functionality.

Table 508. Real-Time Clock register map

Name	Description	Access	Reset Value ^[1]	Address
Miscellaneous registers (see Section 27.6.2)				
ILR	Interrupt Location Register	R/W	0	0x4002 4000
CCR	Clock Control Register	R/W	NC	0x4002 4008
CIIR	Counter Increment Interrupt Register	R/W	0	0x4002 400C
AMR	Alarm Mask Register	R/W	0	0x4002 4010
RTC_AUX	RTC Auxiliary control register	R/W	0x10	0x4002 405C
RTC_AUXEN	RTC Auxiliary Enable register	R/W	0	0x4002 4058
Consolidated time registers (see Section 27.6.3)				
CTIME0	Consolidated Time Register 0	RO	NC	0x4002 4014
CTIME1	Consolidated Time Register 1	RO	NC	0x4002 4018
CTIME2	Consolidated Time Register 2	RO	NC	0x4002 401C
Time counter registers (see Section 27.6.4)				
SEC	Seconds Counter	R/W	NC	0x4002 4020
MIN	Minutes Register	R/W	NC	0x4002 4024
HOURL	Hours Register	R/W	NC	0x4002 4028
DOM	Day of Month Register	R/W	NC	0x4002 402C
DOW	Day of Week Register	R/W	NC	0x4002 4030
DOY	Day of Year Register	R/W	NC	0x4002 4034
MONTH	Months Register	R/W	NC	0x4002 4038
YEAR	Years Register	R/W	NC	0x4002 403C
CALIBRATION	Calibration Value Register	R/W	NC	0x4002 4040
General purpose registers (see Section 27.6.6)				
GPREG0	General Purpose Register 0	R/W	NC	0x4002 4044
GPREG1	General Purpose Register 1	R/W	NC	0x4002 4048
GPREG2	General Purpose Register 2	R/W	NC	0x4002 404C
GPREG3	General Purpose Register 3	R/W	NC	0x4002 4050
GPREG4	General Purpose Register 4	R/W	NC	0x4002 4054
Alarm register group (see Section 27.6.7)				
ALSEC	Alarm value for Seconds	R/W	NC	0x4002 4060
ALMIN	Alarm value for Minutes	R/W	NC	0x4002 4064
ALHOUR	Alarm value for Hours	R/W	NC	0x4002 4068
ALDOM	Alarm value for Day of Month	R/W	NC	0x4002 406C
ALDOW	Alarm value for Day of Week	R/W	NC	0x4002 4070
ALDOY	Alarm value for Day of Year	R/W	NC	0x4002 4074
ALMON	Alarm value for Months	R/W	NC	0x4002 4078
ALYEAR	Alarm value for Year	R/W	NC	0x4002 407C

- [1] Reset values apply only to a power-up of the RTC block, other types of reset have no effect on this block. Since the RTC is powered whenever either of the $V_{DD(REG)(3V3)}$ or V_{BAT} supplies are present, power-up reset occurs only when both supplies were absent and then one is turned on. Most registers are not affected by power-up of the RTC and must be initialized by software if the RTC is enabled. The Reset Value reflects the data stored in used bits only. It does not include reserved bits content.

27.6.1 RTC interrupts

Interrupt generation is controlled through the Interrupt Location Register (ILR), Counter Increment Interrupt Register (CIIR), the alarm registers, and the Alarm Mask Register (AMR). Interrupts are generated only by the transition into the interrupt state. The ILR separately enables CIIR and AMR interrupts. Each bit in CIIR corresponds to one of the time counters. If CIIR is enabled for a particular counter, then every time the counter is incremented an interrupt is generated. The alarm registers allow the user to specify a date and time for an interrupt to be generated. The AMR provides a mechanism to mask alarm compares. If all non-masked alarm registers match the value in their corresponding time counter, then an interrupt is generated.

The RTC interrupt can bring the microcontroller out of Power-down mode when the RTC is operating from its own oscillator on the RTCX1-2 pins. When the RTC interrupt is enabled for wake-up and its selected event occurs, the oscillator wake-up cycle associated with the XTAL1/2 pins is started. For details on the RTC based wake-up process see [Section 4.8.8 “Wake-up from Reduced Power Modes” on page 64](#) and [Section 4.9 “Wake-up timer” on page 67](#).

27.6.2 Miscellaneous register group

27.6.2.1 Interrupt Location Register (ILR - 0x4002 4000)

The Interrupt Location Register is a 2-bit register that specifies which blocks are generating an interrupt (see [Table 509](#)). Writing a one to the appropriate bit clears the corresponding interrupt. Writing a zero has no effect. This allows the programmer to read this register and write back the same value to clear only the interrupt that is detected by the read.

Table 509. Interrupt Location Register (ILR - address 0x4002 4000) bit description

Bit	Symbol	Description	Reset value
0	RTCCIF	When one, the Counter Increment Interrupt block generated an interrupt. Writing a one to this bit location clears the counter increment interrupt.	0
1	RTCALF	When one, the alarm registers generated an interrupt. Writing a one to this bit location clears the alarm interrupt.	0
31:21	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.2.2 Clock Control Register (CCR - 0x4002 4008)

The clock register is a 4-bit register that controls the operation of the clock divide circuit. Each bit of the clock register is described in [Table 510](#). All NC bits in this register should be initialized when the RTC is first turned on.

Table 510. Clock Control Register (CCR - address 0x4002 4008) bit description

Bit	Symbol	Value	Description	Reset value
0	CLKEN		Clock Enable.	NC
		1	The time counters are enabled.	
		0	The time counters are disabled so that they may be initialized.	

Table 510. Clock Control Register (CCR - address 0x4002 4008) bit description ...continued

Bit	Symbol	Value	Description	Reset value
1	CTCRST		CTC Reset.	0
		1	When one, the elements in the internal oscillator divider are reset, and remain reset until CCR[1] is changed to zero. This is the divider that generates the 1 Hz clock from the 32.768 kHz crystal. The state of the divider is not visible to software.	
		0	No effect.	
3:2	-		Internal test mode controls. These bits must be 0 for normal RTC operation.	NC
4	CCALEN		Calibration counter enable.	NC
		1	The calibration counter is disabled and reset to zero.	
		0	The calibration counter is enabled and counting, using the 1 Hz clock. When the calibration counter is equal to the value of the CALIBRATION register, the counter resets and repeats counting up to the value of the CALIBRATION register. See Section 27.6.4.2 and Section 27.6.5 .	
31:5	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.2.3 Counter Increment Interrupt Register (CIIR - 0x4002 400C)

The Counter Increment Interrupt Register (CIIR) gives the ability to generate an interrupt every time a counter is incremented. This interrupt remains valid until cleared by writing a 1 to bit 0 of the Interrupt Location Register (ILR[0]).

Table 511. Counter Increment Interrupt Register (CIIR - address 0x4002 400C) bit description

Bit	Symbol	Description	Reset value
0	IMSEC	When 1, an increment of the Second value generates an interrupt.	0
1	IMMIN	When 1, an increment of the Minute value generates an interrupt.	0
2	IMHOUR	When 1, an increment of the Hour value generates an interrupt.	0
3	IMDOM	When 1, an increment of the Day of Month value generates an interrupt.	0
4	IMDOW	When 1, an increment of the Day of Week value generates an interrupt.	0
5	IMDOY	When 1, an increment of the Day of Year value generates an interrupt.	0
6	IMMON	When 1, an increment of the Month value generates an interrupt.	0
7	IMYEAR	When 1, an increment of the Year value generates an interrupt.	0
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.2.4 Alarm Mask Register (AMR - 0x4002 4010)

The Alarm Mask Register (AMR) allows the user to mask any of the alarm registers. [Table 512](#) shows the relationship between the bits in the AMR and the alarms. For the alarm function, every non-masked alarm register must match the corresponding time counter for an interrupt to be generated. The interrupt is generated only when the counter comparison first changes from no match to match. The interrupt is removed when a one is written to the appropriate bit of the Interrupt Location Register (ILR). If all mask bits are set, then the alarm is disabled.

Table 512. Alarm Mask Register (AMR - address 0x4002 4010) bit description

Bit	Symbol	Description	Reset value
0	AMRSEC	When 1, the Second value is not compared for the alarm.	0
1	AMRMIN	When 1, the Minutes value is not compared for the alarm.	0
2	AMRHOURL	When 1, the Hour value is not compared for the alarm.	0
3	AMRDOM	When 1, the Day of Month value is not compared for the alarm.	0
4	AMRDOW	When 1, the Day of Week value is not compared for the alarm.	0
5	AMRDOY	When 1, the Day of Year value is not compared for the alarm.	0
6	AMRMON	When 1, the Month value is not compared for the alarm.	0
7	AMRYEAR	When 1, the Year value is not compared for the alarm.	0
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.2.5 RTC Auxiliary control register (RTC_AUX - 0x4002 405C)

The RTC Auxiliary Control register contains added interrupt flags for functions that are not part of the Real Time Clock itself (the part recording the passage of time and generating other time related functions). On the LPC176x/5x, the only added interrupt flag is for failure of the RTC oscillator.

Table 513. RTC Auxiliary control register (RTC_AUX - address 0x4002 405C) bit description

Bit	Symbol	Description	Reset value
3:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
4	RTC_OSCF	RTC Oscillator Fail detect flag. RTC_OSCF RTC Oscillator Fail detect flag. Read: This bit is set when the RTC oscillator fails to toggle on the next cycle, and when RTC power is first turned on. An interrupt occurs when this bit is set, if enabled. The RTC_OSCFEN bit in RTC_AUXEN is a 1, and the RTC interrupt is enabled in the NVIC. Write: writing a 1 to this bit clears the flag.	1
31:5	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.2.6 RTC Auxiliary Enable register (RTC_AUXEN - 0x4002 4058)

The RTC Auxiliary Enable Register controls whether additional interrupt sources represented in the RTC Auxiliary control register are enabled.

Table 514. RTC Auxiliary Enable register (RTC_AUXEN - address 0x4002 4058) bit description

Bit	Symbol	Description	Reset value
3:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
4	RTC_OSCFEN	Oscillator Fail Detect interrupt enable. When 0: the RTC Oscillator Fail detect interrupt is disabled. When 1: the RTC Oscillator Fail detect interrupt is enabled. See Section 27.6.2.5 .	0
31:5	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.3 Consolidated time registers

The values of the Time Counters can optionally be read in a consolidated format which allows the programmer to read all time counters with only three read operations. The various registers are packed into 32-bit values as shown in [Table 515](#), [Table 516](#), and [Table 517](#). The least significant bit of each register is read back at bit 0, 8, 16, or 24.

The Consolidated Time Registers are read-only. To write new values to the Time Counters, the Time Counter addresses should be used.

27.6.3.1 Consolidated Time Register 0 (CTIME0 - 0x4002 4014)

The Consolidated Time Register 0 contains the low order time values: Seconds, Minutes, Hours, and Day of Week.

Table 515. Consolidated Time register 0 (CTIME0 - address 0x4002 4014) bit description

Bit	Symbol	Description	Reset value
5:0	Seconds	Seconds value in the range of 0 to 59	NC
7:6	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
13:8	Minutes	Minutes value in the range of 0 to 59	NC
15:14	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
20:16	Hours	Hours value in the range of 0 to 23	NC
23:21	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NC
26:24	Day Of Week	Day of week value in the range of 0 to 6	NA
31:27	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NC

27.6.3.2 Consolidated Time Register 1 (CTIME1 - 0x4002 4018)

The Consolidate Time Register 1 contains the Day of Month, Month, and Year values.

Table 516. Consolidated Time register 1 (CTIME1 - address 0x4002 4018) bit description

Bit	Symbol	Description	Reset value
4:0	Day of Month	Day of month value in the range of 1 to 28, 29, 30, or 31 (depending on the month and whether it is a leap year).	NC
7:5	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
11:8	Month	Month value in the range of 1 to 12.	NC
15:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
27:16	Year	Year value in the range of 0 to 4095.	NC
31:28	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.3.3 Consolidated Time Register 2 (CTIME2 - 0x4002 401C)

The Consolidate Time Register 2 contains just the Day of Year value.

Table 517. Consolidated Time register 2 (CTIME2 - address 0x4002 401C) bit description

Bit	Symbol	Description	Reset value
11:0	Day of Year	Day of year value in the range of 1 to 365 (366 for leap years).	NC
31:12	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.4 Time Counter Group

The time value consists of the eight counters shown in [Table 518](#) and [Table 519](#). These counters can be read or written at the locations shown in [Table 519](#).

Table 518. Time Counter relationships and values

Counter	Size	Enabled by	Minimum value	Maximum value
Second	6	1 Hz Clock	0	59
Minute	6	Second	0	59
Hour	5	Minute	0	23
Day of Month	5	Hour	1	28, 29, 30 or 31
Day of Week	3	Hour	0	6
Day of Year	9	Hour	1	365 or 366 (for leap year)
Month	4	Day of Month	1	12
Year	12	Month or day of Year	0	4095

Table 519. Time Counter registers

Name	Size	Description	Access	Address
SEC	6	Seconds value in the range of 0 to 59	R/W	0x4002 4020
MIN	6	Minutes value in the range of 0 to 59	R/W	0x4002 4024
HOUR	5	Hours value in the range of 0 to 23	R/W	0x4002 4028
DOM	5	Day of month value in the range of 1 to 28, 29, 30, or 31 (depending on the month and whether it is a leap year). ^[1]	R/W	0x4002 402C
DOW	3	Day of week value in the range of 0 to 6 ^[1]	R/W	0x4002 4030
DOY	9	Day of year value in the range of 1 to 365 (366 for leap years) ^[1]	R/W	0x4002 4034
MONTH	4	Month value in the range of 1 to 12	R/W	0x4002 4038
YEAR	12	Year value in the range of 0 to 4095	R/W	0x4002 403C

[1] These values are simply incremented at the appropriate intervals and reset at the defined overflow point. They are not calculated and must be correctly initialized in order to be meaningful.

27.6.4.1 Leap year calculation

The RTC does a simple bit comparison to see if the two lowest order bits of the year counter are zero. If true, then the RTC considers that year a leap year. The RTC considers all years evenly divisible by 4 as leap years. This algorithm is accurate from the year 1901 through the year 2099, but fails for the year 2100, which is not a leap year. The only effect of leap year on the RTC is to alter the length of the month of February for the month, day of month, and year counters.

27.6.4.2 Calibration register (CALIBRATION - address 0x4002 4040)

The following register is used to calibrate the time counter.

Table 520. Calibration register (CALIBRATION - address 0x4002 4040) bit description

Bit	Symbol	Value	Description	Reset value
16:0	CALVAL	-	If enabled, the calibration counter counts up to this value. The maximum value is 131, 072 corresponding to about 36.4 hours. Calibration is disabled if CALVAL = 0.	NC
17	CALDIR		Calibration direction	NC
		1	Backward calibration. When CALVAL is equal to the calibration counter, the RTC timers will stop incrementing for 1 second.	
		0	Forward calibration. When CALVAL is equal to the calibration counter, the RTC timers will jump by 2 seconds.	
31:12			Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

27.6.5 Calibration procedure

The calibration logic can periodically adjust the time counter either by not incrementing the counter, or by incrementing the counter by 2 instead of 1. This allows calibrating the RTC oscillator under some typical voltage and temperature conditions without the need to externally trim the RTC oscillator.

A recommended method for determining the calibration value is to use the CLKOUT feature to unintrusively observe the RTC oscillator frequency under the conditions it is to be trimmed for, and calculating the number of clocks that will be seen before the time is off by one second. That value is used to determine CALVAL.

If the RTC oscillator is trimmed externally, the same method of unintrusively observing the RTC oscillator frequency may be helpful in that process.

Backward calibration

Enable the RTC timer and calibration in the CCR register (set bits CLKEN = 1 and CCALEN = 0). In the CALIBRATION register, set the calibration value CALVAL ≥ 1 and select CALDIR = 1.

- The SEC timer and the calibration counter count up for every 1 Hz clock cycle.
- When the calibration counter reaches CALVAL, a calibration match occurs and all RTC timers will be stopped for one clock cycle so that the timers will not increment in the next cycle.
- If an alarm match event occurs in the same cycle as the calibration match, the alarm interrupt will be delayed by one cycle to avoid a double alarm interrupt.

Forward calibration

Enable the RTC timer and calibration in the CCR register (set bits CLKEN = 1 and CCALEN = 0). In the CALIBRATION register, set the calibration value CALVAL ≥ 1 and select CALDIR = 0.

- The SEC timer and the calibration counter count up for every 1 Hz clock cycle.
- When the calibration counter reaches CALVAL, a calibration match occurs and the RTC timers are incremented by 2.
- When the calibration event occurs, the LSB of the ALSEC register is forced to be one so that the alarm interrupt will not be missed when skipping a second.

27.6.6 General purpose registers

27.6.6.1 General purpose registers 0 to 4 (GPREG0 to GPREG4 - addresses 0x4002 4044 to 0x4002 4054)

These registers can be used to store important information when the main power supply is off. The value in these registers is not affected by chip reset.

Table 521. General purpose registers 0 to 4 (GPREG0 to GPREG4 - addresses 0x4002 4044 to 0x4002 4054) bit description

Bit	Symbol	Description	Reset value
31:0	GP0 to GP4	General purpose storage.	N/A

27.6.7 Alarm register group

The alarm registers are shown in [Table 522](#). The values in these registers are compared with the time counters. If all the unmasked (See [Section 27.6.2.4 “Alarm Mask Register \(AMR - 0x4002 4010\)” on page 574](#)) alarm registers match their corresponding time counters then an interrupt is generated. The interrupt is cleared when a 1 is written to bit 1 of the Interrupt Location Register (ILR[1]).

Table 522. Alarm registers

Name	Size	Description	Access	Address
ALSEC	6	Alarm value for Seconds	R/W	0x4002 4060
ALMIN	6	Alarm value for Minutes	R/W	0x4002 4064
ALHOUR	5	Alarm value for Hours	R/W	0x4002 4068
ALDOM	5	Alarm value for Day of Month	R/W	0x4002 406C
ALDOW	3	Alarm value for Day of Week	R/W	0x4002 4070
ALDOY	9	Alarm value for Day of Year	R/W	0x4002 4074
ALMON	4	Alarm value for Months	R/W	0x4002 4078
ALYEAR	12	Alarm value for Years	R/W	0x4002 407C

27.7 RTC usage notes

If the RTC is used, V_{BAT} may be connected to an independent power supply (typically an external battery), or left floating. The RTC domain will always be internally powered if $V_{DD(REG)(3V3)}$ is present, even if there is no power applied to V_{BAT} . As long as power is available on either $V_{DD(REG)(3V3)}$ or V_{BAT} , the RTC will retain its time value and the backup register contents. If both $V_{DD(REG)(3V3)}$ and V_{BAT} are not present, all RTC information will be lost. RTC incrementation will stop or be unpredictable if the clock source is lost, interrupted, or altered.